Detecting Gait Anomalies Based on the Analysis of Foot Pressures

Paweł Zawadzki
Institute of Theory of Electrical Engineering, Measurement and Information Systems
Warsaw University of Technology
Warsaw, Poland
zawadzp4@ee.pw.edu.pl

Jacek Starzyński
Institute of Theory of Electrical Engineering, Measurement and Information Systems
Warsaw University of Technology
Warsaw, Poland
jstar@ee.pw.edu.pl

***Abstract*—The modern professional medical devices for gait analysis are very advanced but are not portable and available everywhere. However, the current information about our gait ("permanent" monitoring) might be helpful to improve our health condition.**

This paper describes the idea of gait anomalies detection by algorithms of deep learning. These algorithms analyze data collected by a prototype measuring device managed by a mobile application. Sensors collect foot pressure values as data for deep learning. The problem of collecting real data (from real patients) is discussed. The potential use of the device described is also discussed.

Finally, assumptions and expectations regarding the results were compared with several related projects of deep learning and gait analysis. Our analysis of related studies shows that the use of deep neural networks (convolutional and recurrent) can have a satisfactory probability of detecting gait anomalies.

***Index Terms*—gait anomalies detection, pressure sensors, insole, Arduino, mobile application, deep learning**

I. Introduction

The incredible development of science and scientific knowledge has made the opportunity to break many obstacles in our daily life. People have invented many groundbreaking tools to make life easier but still for now using legs in the process of movement is the most basic, natural and dominant method. This is the reason why we should care more about our overall health and quality of life.

One of the many ideas is to detect the problem before much more complex symptoms appear. It is important to note that walking abnormalities can be a source of health problems so we should detect those abnormalities to prevent negative health effects [1], [2].

Our idea is based on the analysis of the value of pressure sensors mounted under the insole of the shoe. Not exactly the pressure value – it is much more important to monitor the pressure changes over time. If we collect a lot of data sets from many people with and without gait abnormalities, we will be able to start training a deep neural network to identify normal and abnormal gait. What is more, the result of this process may be the first step to early detection of a potential health risk (from the point of view of walking).

This paper is about research planning in the field of detection of abnormal gait.

II. Description of measuring device

Before starting any complex data analysis, we need a device to collect data.

The prototype of the measuring device was made of several ready-to-use electrical components. The core of this device is Arduino MKR Zero (48 MHz, clock 32 KB SRAM, 256 KB flash memory). The board is powered by two 18650 batteries connected in series, resulting in a voltage between 7 V and 9 V. The resulting output voltage is too high for Arduino MKR Zero, which expected to be exactly equal 5 V. This is the reason to use a step-down converter (D24V5F5), which gives exactly 5 V at the output.

To collect measurement data, pressure sensors are needed. In this device, we use six identical pressure sensors FSR-402 [3]. These sensors are very sensitive, but they work correctly only in a narrow range (up to several kilograms). This narrow range can be the biggest disadvantage of our prototype. In fact, we assumed that the most important value from our point of view would be to monitor changes in pressure over time. This means that we do not need to know the exact value of pressure over time. In this approach, it will be much more important to record pressure changes in real time. Accuracy of pressure measurements will be sufficient to determine which part of the foot touches the ground.

The built prototype was designed to use many identical pressure sensors, but we would like to use as few sensors as possible. The current version of the device has only six sensors (three sensors per insole).

The Arduino MKR Zero has only seven analog input pins. We have connected all the sensors to the multiplexer (74HC4051), and the multiplexer is directly connected to the Arduino board to limit the use of these input pins.

It should be noted that the location of the sensors on the insole is not random which means that the sensors are located under selected bones of the foot: first metatarsal bone, fifth metatarsal bone, calcaneus (heel bone) [4].

To improve the comfort of using the prototype, the Bluetooth Low Energy module (HM-10) has been connected to Arduino. Thanks to this, the managing of measurement (starting, stopping) could be made remotely by mobile application.

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The last important component of the described device is the buzzer for communicating the actual state of the device (busy, standby mode, power-off, memory card error).

All described elements are enclosed in a plastic casing (Fig. 1) designed in an easy-to-use CAD application [5] and printed on a Prusa i3 MK3 3D printer.

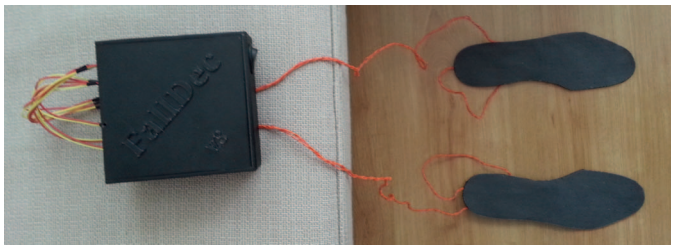

Fig. 1. On the left: the plastic casing (44x110x115 mm) with all electrical components. On the right: two insoles for shoes with pressure sensors connected by wires.

III. The course of research

A. Preparing a patient (or a volunteer) for measuring the pressure of the feet

Before starting the measurement the patient should put the device (Fig. 1) on himself. At first, he should put the insoles into the shoes. Next, he should attach the device (the part with plastic housing) to the belt.

After this short preparation, we can start the device using the button on the housing. A repeated quick beep produced by the buzzer indicates that the device is ready for use.

The last step to start the process of collecting pressure data sets is to choose the right menu option in the mobile application. Then the data will be collected and saved on the memory card in real time and we will be able to display the preview of the current data in the mobile application (Fig. 2).

B. Data sources

Patients with and without anomalies are data sources. Thus, the main goal in the first phase of the study is to collect data from ”healthy” patients (without defined gait anomalies) and ”unhealthy” (with specific gait anomalies).

Data should be collected from many people during their normal daily activities. We assume that data will be collected only during physical activity.

In order to get better results in the first phase of testing algorithms for the detection of gait anomalies, it is worth taking the effort (when collecting data) to mark the type of activities such as going up stairs, going down stairs, walking on steep ground, walking on even ground, running.

Another type of useful information that can be saved during data collection is patient’s data, such as age, sex, weight, and height, because the comparison of data of significantly different people may cause interpretation problems of the obtained results.

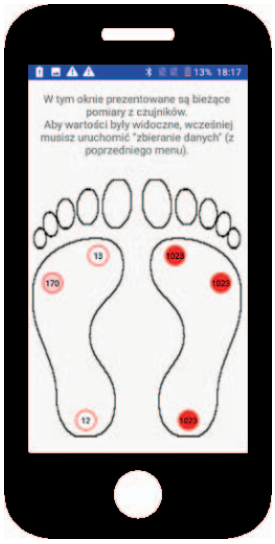

Fig. 2. Example of live preview. Higher saturation with red color indicates greater pressure force. The entire weight was placed on the right foot. At the time of measurement, the left foot did not touch the ground.

C. Recognizing patterns of gait anomalies

The main purpose of the described research is to build a deep neural network to detect gait distortion. Ideally, to detect the type of distortion, not just detect the fact of deviation from correct gait. From this point of view, the biggest challenge is to collect data from at least a few patients with the same cause of gait distortion.

Suppose we have collected data from ten groups of patients with the same cause of gait distortion. The next step is to build a neural network with architecture closely related to the number of collected data.

At present, machine learning is a rapidly growing field of computer science. Part of this area is deep learning [6] with the popular type of deep neural networks, such as CNN (convolutional neural network) – mainly for images and RNN (recursive neural network) – for processing data collected over a period of time [7]. We would like to use both types of networks independently and then compare the results obtained between them.

Long short-term memory (LSTM) is a very popular [8], [9] type of RNN with surprisingly good learning results and that is why we will try to use this architecture to detect a pattern of gait anomalies. LSTM consists of four important elements: cell, input gate, forget gate, output gate. Connections between those three gates make the possibility to forget about unimportant data (e.g. data collected during standing without moving), strengthening valuable data and improving generalization ability for recognizing gait pattern.

Another mentioned approach uses a convolutional neural network (CNN), which consists of a sequence of several typical layers, e.g. a convolution layer, a pooling layer, a fully-connected layer. Connections between successive layers cause the discovery of features in the data, in our case, they could

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be the features of the gait disorders. The last (fully-connected) layer has a dimension equal to the number of patient groups (e.g. 10 – as we assumed in the first part of this section).

D. Preparation of data for a deep learning network

The prototype pressure measuring device has six pressure sensors. In other words, six pressure values are collected at the same time (L_1, L_2, L_3 – 3 sensors from the left insole; R_1, R_2, R_3 – 3 sensors from the right insole).

$$X_{i,t} = (L_1, L_2, L_3, R_1, R_2, R_3) \tag{1}$$

the t-th measurement in time from the i-th person. This is illustrated in the figure 3.

The CNN variant has an analogous structure. All i-th patient data is a single set of input data for the network.

The next layers will be convolution and pooling repeated 2 or 3 times (depending on the results). The last layer should be a gait distortion classifier.

E. Potential application of results

We expect that the neural networks from the previous section (III-C) will be sufficient to recognize ”normal” and ”abnormal” gait.

Thus, we can find several possible applications of the described device.

1) A diagnosis that the patient’s gait has some distortion: Suppose a device is really cheap if it is true, many people around the world can buy it and use it (as naturally as using the bathroom scale) in a routine health examination.

Another way is to extend the scope of the basic medical examination with gait analysis as part of the medical condition examination. The main advantage of this solution is the diagnosis of gait distortion at the possibly initial level, which is easier to cure. This can be very helpful for both older and younger people because the comfort of walking is closely related to the comfort of living [10].

2) For the patients after surgery (e.g. hip arthroplasty surgery): People after complicated surgeries might have a problem to get back to full health due to an insufficient number of rehabilitation hours [2]. The device with the taught neural network could be a kind of electrical physiotherapist who will be motivating the patient to correct his gait on daily basis. In effect, it could help in achieving better results of physiotherapy.

3) Prediction and diagnosis of defined gait disorders: Suppose, as assumed in section III-C, that we have collected data from 10 groups of patients. Patients with the same gait anomalies belong to individual groups. If we accumulate enough data and properly train the neural network in recognizing specific gait distortion based on the collected data, the device will help people to a much wider scope in the early diagnosis of gait anomalies because we will try to prepare device (more precisely, we will try to teach the neural network) to predict possible patient’s gait distortion which may occur in the nearest future.

IV. Discussion about the idea

All previous sections omit problematic cases of the described idea. Unfortunately, there are many of them.

A. Advantages and disadvantages

First of all, artificial neural networks (including deep networks) for proper work in order to obtain a good generalization requires a large set of data. In our described case the training data for neural network come from patients (or volunteers) which means that we need a lot of patients with diagnosed gait anomalies. It should be underscored that the consent of the patient for collecting data is required. Not every patient wants to take part in the study. That is why collecting data from many people is truly harder than it looks.

Moreover, if we realize that we need sets of data of each group of patients with diagnosed gait distortion, the task of data collection becomes more logistically complex. This is one of the reasons why building a neural network to recognize an ”abnormal” gait pattern (due to insufficient rehabilitation after surgery or as a result of the body’s compensatory mechanisms in reaction to pain or having other bases of complications) is more likely to be done.

It is intuitively perceptible that classification between two classes (“normal” gait and “abnormal” gait) is easier to perform with a better level of correct classifications than classification between multiple classes (recognizing the type of gait abnormalities).

Another doubt closely related to the previous one is the number of pressure sensors attached to a single insole. We expect that the pressure values aggregated from six sensors (from two insoles) will be sufficient to effectively meet the basic expectations regarding the gait classification (recognize the ”normal” and ”abnormal” pattern). We suppose the problem will occur with much more difficult classification for recognizing the type of gait distortion. In fact, we are not sure the number of used pressure sensors.

On the other hand, our basic assumption was to minimize the number of sensors. We decided arbitrarily about the number of sensors used. And there is no heuristic background in this decision.

B. Comparison with other ideas

From the late 20th century to the present, many scientists have tried to monitor and analyze human gait – they had different motivations for research.

At present, there are a few popular types of systems for monitoring human gait. One of them is a specialist laboratory with infrared cameras and floor with mounted pressure sensors. Based on the recorded video from the camera and measurement data from floor mounted force sensors the scientists try to determine if a patient’s gait is correct [11], [12].

Another popular approach is a specialist treadmill that creates detailed visualization of a patient’s gait [13]. The gait model thus prepared can be used to diagnose postural defects (associated with distortion of walking) and perform much more effective physiotherapy.

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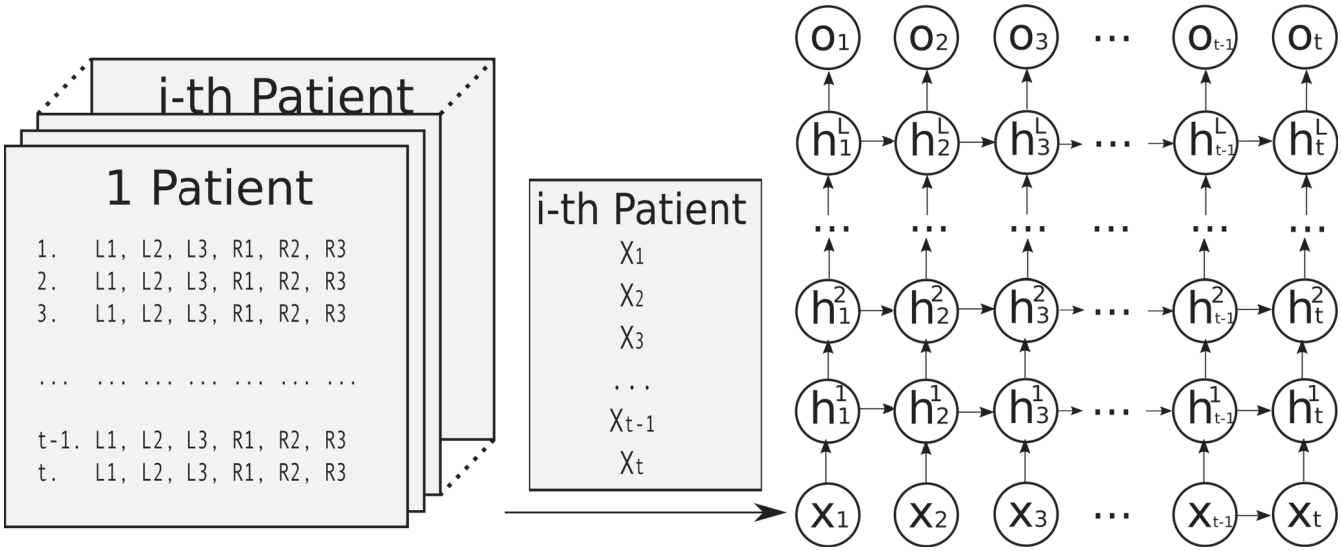

Fig. 3. Graphical representation of collected data for LSTM. Each patient collects T data sets. Each data set contains values from six pressure sensors. All data sets of i-th patient are passed to the LSTM input layer. L hidden layers (h^L) connected to each other processes data into the output layer (O). Important note: the classification layer (e.g. softmax layer) is not visible in the LSTM scheme.

The newest idea which is created from the transformation of MIT's project is a specialist mat that performs a one-minute test of health condition [14]. This project uses machine learning algorithms to detect balance changing and evaluate to fall risk in the near future.

An idea unrelated to walking distortion, but with gait analysis in order to identify people, is based on data collected from smartphone sensors, such as accelerometer and gyroscope [15]. These data are sent to two neural networks (LSTM and CNN). Obtained correct results are higher than 93%. These networks used six measurement values (two sensors with three coordinates).

This project is very similar to our idea and might confirm the thesis about a sufficient number of sensors.

V. Conclusion

Gait analysis using deep learning algorithms is not an obvious way to detect walking abnormalities. However, the area of research related to gait analysis is still being studied by scientists [16].

We introduced the idea of detecting gait anomalies based on the analysis of data collected by pressure sensors in the shoe insoles. Our concept, as other studies show, has a chance of success, but the biggest challenge from our point of view is to collect accurate training data for RNN and CNN from real patients.

Detecting gait anomalies (in the first approach ignoring the detection of anomaly type) is the goal of further research.

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